# POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2017
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

बालकुमारी, ललितपुर ९८०-१५९९५९२
NCIT College

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| 1. | a) | "If computer passes Turing Test, Computer acts like human". Do you agree? State your answer with reason. | | 7 |
| | b) | Define rational agent? Explain the Learning agent with block diagram. | | 8 |
| 2. | a) | Solve the following problem as CSP.<br>LOGIC + LOGIC= PROLOG | | 7 |
| | b) | Differentiate informed and blind search. How DFS is different from BFS? Compare it with evaluation parameters. | | 8 |
| 3. | a) | Write down the disadvantages of hill climbing search procedure. When does simulated annealing algorithm behave like hill climbing? | | 7 |
| | b) | Prove 'Vinod buys a ticket' using resolution considering following premises - Everyone who enters in a theatre has to buy a ticket. Person who doesn't have money can't buy a ticket. Vinod enter a theatre. | | 8 |
| 4. | a) | Convert the following statement to clause form and discuss the steps.<br>$\forall x[B(x) \rightarrow (\exists y [Q(x,y) \wedge \neg P(y)]$ | | 8 |
| | b) | What do you mean by learning by analogy? Explain derivational analogy. | | 7 |
| 5. | a) | "Describe briefly about Nearest Neighbour and Support Vector Machine Learning methods. | | 8 |
| | b) | What is machine learning? In computers, in many cases, rote learning is used. Give five such examples of rote learning. | | 7 |
| 6. | a) | What is Artificial Neural Network? Implement a simple logical OR using the perceptron model. What are the different possible threshold values? | | 8 |
| | b) | Explain the steps involved in Natural Language Processing. | | 7 |
| 7. | Write short notes on: **(Any two)** | | | 2× |



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a) Fuzzy Learning
b) Expert System
c) Probability and Baye's Theorem

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